

Muscle Cooling Before and in the Middle of a Session: There Are Benefits on Subsequent Localized Endurance Performance in a Warm Environment

Jiří Baláš, Jan Kodejška, Adéla Procházková, Roman Knap, and James J. Tufano

Faculty of Physical Education and Sport, Charles University, Prague, Czech Republic

Abstract

Baláš, J, Kodejška, J, Procházková, A, Knap, R, and Tufano, JJ. Muscle cooling before and in the middle of a session: there are benefits on subsequent localized endurance performance in a warm environment. *J Strength Cond Res* XX(X): 000–000, 2023—Localized cold-water immersion (CWI) has been shown to facilitate recovery in the middle of a session of exhaustive repeated forearm contractions. However, it has been suggested that these benefits may be attributed to “precooling” the muscle before an activity, as opposed to cooling a previously overheated muscle. Therefore, this study aimed to determine how precooling and mid-cooling affects localized repeated muscular endurance performance in a warm environment. Nineteen subjects completed a familiarization session and 3 laboratory visits, each including 2 exhaustive climbing trials separated by 20 minutes of recovery: PRE_{CWI} (CWI, trial 1; passive sitting [PAS], trial 2); MID_{CWI} (PAS, trial 1; CWI, trial 2); and CONTROL (PAS, trial 1; PAS, trial 2). Climbing trial 1 in PRE_{CWI} was 32 seconds longer than in CONTROL ($p = 0.013$; $d = 0.46$) and 47 seconds longer than in MID_{CWI} ($p = 0.001$; $d = 0.81$). The time of climbing trial 2 after PAS (PRE_{CWI} and CONTROL) was very similar (312 vs. 319 seconds) irrespective of the first trial condition. However, the time of the second trial in MID_{CWI} was 43 seconds longer than in PRE_{CWI} ($p < 0.001$; $d = 0.63$) and 50 seconds longer than in CONTROL ($p < 0.001$; $d = 0.69$). In warm environments, muscle precooling and mid-cooling can prolong localized endurance performance during climbing. However, the effectiveness of mid-cooling may not be as a “recovery strategy” but as a “precooling” strategy to decrease muscle temperature before subsequent performance, delaying the onset of localized heat-induced neuromuscular fatigue.

Key Words: reliability, cold-water immersion, climbing, precooling, mid-cooling, recovery

Introduction

Muscle cooling using cold-water immersion (CWI) and cooling or ice packs can help maintain performance in hot environments, reduce muscle soreness, and aid recovery from secondary muscle damage (3). It has been suggested that the mechanism by which CWI facilitates these types of acute recovery include ameliorating hyperthermia-mediated fatigue, reducing cardiovascular strain through the redistribution of blood flow from the periphery to the core, and removing muscle metabolites at a quicker rate (15). However, the positive effects of CWI are largely known to be valid for whole-body endurance exercise in hot and humid environments where the metabolic demand of exercise is quite high. As such, the aforementioned mechanisms may not necessarily be as important during recovery from resistance exercise where the metabolic load would likely be lower (20). Reducing metabolic demands even further, localized contractions of small muscle groups such as the finger and wrist flexors likely do not result in hyperthermia, even in hot environments. Yet, it has been shown that muscle cooling can maintain or even increase repeated performance during localized forearm contractions (5,13,16). Therefore, it seems as if localized cooling may also be beneficial for performance, even in the absence of large physiological demands in extreme environments.

Compared with whole-body cooling, the physiological mechanisms responsible for the benefits observed after muscle cooling at the localized level are less understood (20). Logically, the beneficial effects of localized cooling are likely not because of central mechanisms but rather because of peripheral mechanisms. Among these possible peripheral mechanisms, glycolysis and glycogenolysis sparing and gross efficiency have been proposed as the most plausible (8,10,12,28). However, these effects are mostly seen at muscle temperatures of $\sim 27^{\circ}\text{C}$ during repeated exhaustive contractions, whereas lower and higher muscle temperatures lead to progressively deteriorating performances (8). Therefore, exercising in appropriate cool ambient air temperature may optimize muscle temperature and muscle metabolic functions.

Along these lines of cool environments and thereby cooler muscle temperatures, it is well known that the fastest endurance runs are recorded with lower ambient temperatures around $5\text{--}10^{\circ}\text{C}$ (11), whereas progressively slower times that occur in higher temperatures are related to central factor-mediated fatigue. Interestingly, previous research has shown that even sport climbing performance, which relies dominantly on localized finger flexor endurance capacity, was greater at an ambient temperature of 10°C than during a thermoneutral condition (24°C) (18), showing that localized mechanisms are responsible for performance enhancement with respect to thermoneutral conditions as well. Consequently, optimal muscle temperature along with central factors seem to contribute to enhance both whole-body or localized

Address correspondence to Jiří Baláš, balas@ftvs.cuni.cz.

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endurance performance. Consequently, muscle precooling, which is recognized for reducing the impact of hyperthermia on subsequent performance (3,7), may have a beneficial effect (by delaying the onset of subsequent fatigue) for short exhaustive endurance tasks in the heat in sport with localized fatigue, such as during sport climbing, combat sports, or canoe-slalom.

Therefore, an intriguing question presents itself. If muscle cooling only facilitates acute recovery, a subsequent performance should demonstrate minimal decreases, but there should be no performance enhancement. However, if there is a performance enhancement above what would occur at a thermoneutral condition, the muscle cooling must have worked as a type of stimulation method similar to precooling, and the effect on acute recovery may be questionable. Therefore, the aim of this study was to determine to what extent precooling and mid-cooling affects repeated localized endurance performance in the heat. We hypothesized that precooling would enhance subsequent performance in a similar fashion that mid-cooling would improve subsequent performance.

Methods

Experimental Approach to the Problem

Subjects visited the laboratory 4 times. During the first visit, subjects completed the entrance questionnaire, provided their written informed consent, had their body mass and height measured, and were familiarized with the climbing ergometer (treadwall). The treadwall is positioned indoors and enables a participant to climb on a preset route at a given speed and inclination, enabling standardization and replicability of the climbing route. To conclude the familiarization session, 1 trial of the testing protocol was performed with progressively increasing difficulty.

During the next 3 visits, subjects performed 2 exhaustive climbing trials separated by 20 minutes of passive sitting (PAS) or CWI. Each visit contained one of the following conditions, which were completed in a quasi-randomized order: (a) CONTROL—(PAS, trial 1; PAS, trial 2); (b) precooling—PRE_{CWI} (CWI, trial 1; PAS, trial 2); and (c) mid-cooling—MID_{CWI} (PAS, trial 1; CWI, trial 2; Figure 1).

Subjects

Nineteen intermediate-to-advanced male sport climbers (25 ± 6 years range 19–44 years, 74 ± 8 kg, 178 ± 7 cm, 11–23 International Rock-Climbing Association [IRCRA] scale, 6a–8a French/Sport scale) volunteered to participate in the study. All subjects were healthy nonsmokers and were asked to restrain from heavy physical activity and alcohol for 24 hours, caffeine for 12 hours, and heavy food for 2 hours before each test session. Each subject completed all sessions around the same time of the day. The study conformed to the recommendations of the World Medical Association and the Declaration of Helsinki and was approved by the Ethics Committee of Charles University, Faculty of Physical Education and Sport under the No EK 083/2022. All subjects gave their written, informed consent and were allowed to leave the experiment at any stage. Sample size calculations were determined using G*Power (version 3.1.9.7) based on the outcome measures from previous research (2,16). With power ($1 - \beta$ error) set at 0.95, α -level set at 0.05, and an effect size (eta squared) of 0.15 for differences in performance after CWI, the sample size was set as 15 participants. Because of possible dropout and technical errors, 19 participants were recruited.

Procedures

Precooling and Mid-cooling. Localized CWI of both forearms was used for both precooling and mid-cooling, after the CWI protocol that had previously demonstrated a positive impact on handgrip endurance (2). Subjects sat with their knees and hips at 90° flexion, with both arms placed at 20° shoulder abduction and 15° flexion, the elbow at 80° flexion, and slight wrist pronation. The forearms, from the elbow to mid-palm, were immersed in 15° C water. Each CWI protocol consisted of three 6-minute cycles (4 minutes of immersion and 2 minute out of water) and 2 minutes for following performance preparation (drying forearms, putting climbing shoes, chalking fingers and hands, and getting on starting holds positions). Water temperature was regulated by adding ice cubes throughout the whole procedure to maintain a water temperature within $\pm 1^\circ$ C of 15° C. Throughout the CWI protocol, the water was agitated to avoid water heating at the skin's surface. During the passive sitting periods, climbers sat in the same anatomical position but without water in the bucket.

Ambient Air, Water, and Skin Temperatures. The ambient air temperature of the room was kept at $30^\circ \text{C} \pm 1^\circ \text{C}$ using a convective heater. The humidity ranged between 58 and 66%. Water and ambient temperatures were measured by thermistor sensors (Comet system, Rožnov pod Radhoštěm, Czech Republic). Skin temperature was assessed immediately before and after the climbing trials using noncontact infrared thermometer (BC 07; Body Comfort, Kněžves, Czech Republic). When CWI as precooling strategy was applied, skin temperature was assessed even before CWI. The skin temperature was taken at the proximal third of frontal part of forearm and from temporal lobe 2–5 cm away from the skin. The mean temperature from 2 readings was recorded.

Climbing Performance. Exhaustive climbing trials were conducted on a motorized treadwall 3.8 m high with a 6-m long belt (ClimbStation generation, Forssa, Finland) following a previously described protocol (1). To reduce the technical difficulty (allowing fatigue to be the determinant of failure), the route was technically very simple and consisted of 14 hand moves repeated within each belt cycle. The route was graded as an 8 on the IRCRA scale (4+ Sport grade) at the starting wall angle and was marked using the same contrast color of handholds and footholds. During all trials, a speed of $9 \text{ m} \cdot \text{min}^{-1}$ was applied to minimize the opportunity for static resting during the climbs. The incremental test started at 0°, and after each minute, the belt was stopped for 10 seconds to allow climbers dry their hands with chalk. The angle was then decreased by -5° to become progressively negative (overhanging), therefore requiring progressively greater finger flexor strength involvement. Climbers were not allowed to touch the ground during rest periods and were verbally encouraged to climb for as long as possible. Each test ended when a climber reached volitional exhaustion and stepped off of the wall and onto the safety mattress. The time of the total climb was used for analysis. After each climbing trial, climbers reported their perceived exertion on a 6–20 scale (4).

Statistical Analyses

Descriptive statistics (mean $\pm$ SD) were used to characterize anthropometric data, climbing time, and skin temperatures. All climbing times were normally distributed, homoscedastic, and of

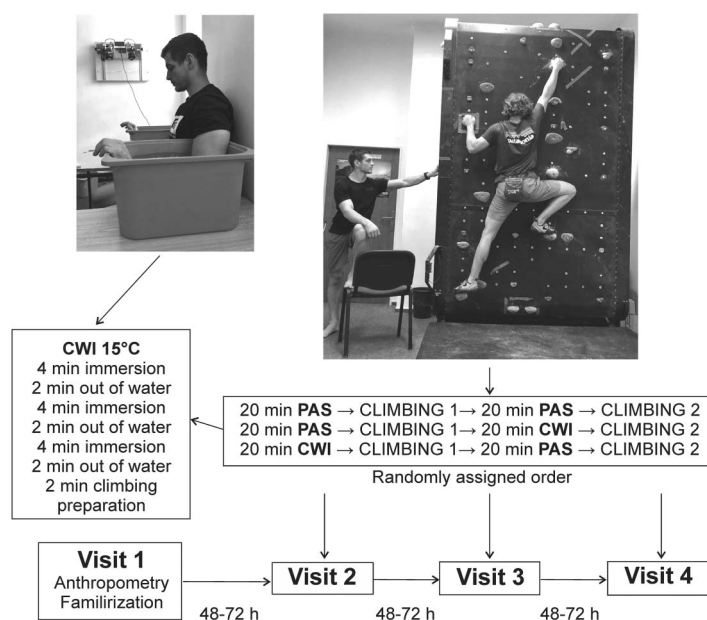


Figure 1. Design of the study. CWI = localized forearm cold-water immersion; PAS = passive recovery; CLIMBING = exhaustive climbing with progressive increase of wall angle.

equal variance. Perceived exertion scores did not meet criteria for normality, so nonparametric tests were performed. Test-retest reliability of climbing performance was calculated from CONTROL and MID_{CWI} visits using only the first trials without any effect of CWI. Intraclass coefficient of reliability (ICC), coefficient of variability (CV), and standard error of measurement were calculated (SEM) (25).

The effects of precooling and mid-cooling on performance were assessed using 2×3 repeated-measures analysis of variance (ANOVA; trial $\times$ condition), and if main effects were significant, a series of paired *t*-tests with Bonferroni corrections were calculated when appropriate. Significance of rating of perceived exertion differences was calculated using Friedman's tests. The significance level was set to 0.05 with 95% confidence intervals (95% CI). Cohen's *d* was calculated to assess the effect size. Microsoft Excel and IBM SPSS 24.0 (IBM, Armonk, NY) were used for all calculations.

Results

Of the 19 climbers, 1 was unable to complete the second trial from 1 session and was included only in the reliability but omitted from the subsequent experimental analyses. The test-retest reliability was satisfactory (test: 330 ± 62 seconds; retest: 345 ± 65 seconds, ICC = 0.90; CV = 6.1%; SEM = 20.5 seconds).

The repeated-measures ANOVA showed a significant main effect of trial ($F_{(1,17)} = 23.5$; $p \leq 0.001$), condition ($F_{(2,34)} = 10.1$; $p \leq 0.001$), and a trial $\times$ condition interaction ($F_{(2,34)} = 34.5$; $p \leq 0.001$). The first climbing trial in PRE_{CWI} was 32 seconds longer than in CONTROL ($p = 0.013$; +9%; $d = 0.46$; 95% CI = 6.4–58.5 seconds) and 47 seconds longer than in MID_{CWI} ($p = 0.001$; +14%; $d = 0.81$; 95% CI = 18.7–75.3 seconds; Figure 2).

After the PAS recovery between trials, there was a significant decrease of performance with greater decrease in the PRE_{CWI} (–62 seconds; –16%; $p < 0.001$; $d = 0.79$; 95% CI = 43.4–80.0 seconds) than in CONTROL (–21 seconds; –6%; $p = 0.002$;

$d = 0.37$; 95% CI = 8.7–33.9 seconds) condition (Figure 3B,C). However, there was a 14-second significant performance increase after cooling in MID_{CWI} ($p = 0.038$; +4%; $d = 0.20$; 95% CI = 0.9–27.5 seconds; Figure 3A).

The average time of the second climbing trials after PAS (PRE_{CWI} and CONTROL) was very similar (312 vs. 319 seconds) irrespective of the first trial condition (Figure 2). However, the time of the second trial in MID_{CWI} was 43 seconds longer than in PRE_{CWI} ($p < 0.001$; +14%; $d = 0.63$; 95% CI = 23.3–63.6 seconds) and 50 seconds longer than in CONTROL ($p < 0.001$; +16%; $d = 0.69$; 95% CI = 25.3–74.7 seconds). Interestingly, the average time from trials 1 and 2 was similar between PRE_{CWI} and MID_{CWI} (350 vs. 355 seconds, $p > 0.05$), whereas the time in CONTROL was significantly shorter (323 seconds, $p < 0.05$).

After PAS, the forearm temperature ranged between ~ 34 and 35°C . The temperature dropped to $23.1^\circ\text{C} \pm 1.7^\circ\text{C}$ and $26.9^\circ\text{C} \pm 4.2^\circ\text{C}$ after cooling in PRE_{CWI} and MID_{CWI}, respectively (Figure 3). The temporal temperature remained stable across all measurements and varied from 36.4 to 37.0°C . Perceived exertion ranged from 17.6 to 18.4 and did not differ ($p > 0.05$) between conditions.

Discussion

This study clearly demonstrated the beneficial effect of localized forearm precooling and mid-cooling on sport climbing endurance performance in a warm controlled environment. Specifically, 20 minutes of localized precooling increased the subsequent performance by ~ 9 –14% with respect to non-cooling conditions; 20 minutes of mid-cooling extended the time of a second exhaustive climb by ~ 14 –16% with respect to the second trials after PAS. Additionally, the effect of precooling lasted only during the first trial and was nonexistent after PAS during the second trial when the forearm skin temperature returned to pre-CWI levels.

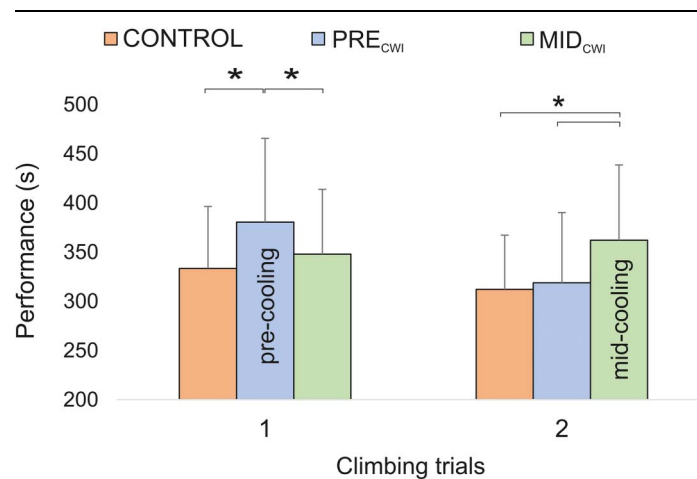


Figure 2. The performance (climbing time in seconds) of 2 repeated trials in 3 conditions. (1) CONTROL: 20 minutes of passive sitting (PAS) before the first trial and the second trial, respectively; (2) precooling (PRE_{CWI}): forearm cold-water immersion (CWI) before the first trial and PAS before the second trial, respectively; and (3) mid-cooling (MID_{CWI}): PAS before the first trial and 20 minutes of CWI before the second trial. *Significant time differences between conditions.

It is well established that CWI undertaken before exercise enhances aerobic performance in the heat (3,7), although the physiological mechanisms are complex and probably very different for localized and whole-body exercise. As previous research has shown that core temperature is not directly related to performance changes (3,7), Cheuvront et al. (6) argued that cardiovascular adjustments accompanying high skin temperature, or in combination with high core temperatures, are the primary mechanisms for performance deterioration as they reduce $\dot{V}O_{2\max}$ and lead to higher relative exercise intensity and perceived exertion. In the current study, the mean performance increase because of precooling was 32 or 47 seconds, which is far above the reliability error. As the perceived exertion and temporal temperature remained similar to PAS and forearm skin temperature did not increase significantly after climbing, it is evident that systemic mechanisms did not lead to a performance increase. Moreover, climbers attain ~85% of their $\dot{V}O_{2\max}$ (1) during similar climbing tests to exhaustion, which supports the assumption that systemic oxygen uptake may not limit climbing performance to exhaustion even in the heat, and localized factors are responsible for premature performance termination.

At a localized level, heat stress during exercise has been shown to impair muscle perfusion and increase the rate of glycogenolysis (12,17). Hence, muscle precooling may nullify the adverse effect of heat on glycogenolysis (10,17). This may be extremely important during repeated isometric contractions, where the oxygen delivery is hindered and the reliance on nonmitochondrial adenosine triphosphate turnover is elevated. For instance, previous research has shown a 38 and 32% increase of force impulse during intermittent handgrip contractions after 15° C and 8° C CWI, respectively, even after previous exhaustive exercise (16). Similarly, a 9% higher force impulse during 60 exhaustive handgrip contractions was demonstrated in subjects after only 6 minutes of localized cryostimulation when compared with controls (9). Therefore, the beneficial effects of cooling appear both in fatigued and in nonfatigued muscles. In contrast, a very low level muscle temperature (less than ~27° C) may slow enzymatic processes

and nerve conduction, therefore athletic performance (muscle endurance, rate of force development, and maximal strength) may be deteriorated (8,10). Considering the above, it is likely that a certain amount of muscle cooling would be optimal, but this remains to be investigated.

In the current study, the performance increase after precooling was very similar to mid-cooling (47 vs. 50 seconds) with respect to the CONTROL condition. Consequently, the total climbing time (trial 1 + 2) was similar in both cooling conditions (MID_{CWI} and PRE_{CWI}) and was longer than in CONTROL. Interestingly, there was a significant increase (4%, 14 seconds) after mid-cooling with respect to trial 1, which is, however, less than the reliability error ($SEM = 20.5$ seconds). However, many individuals prolonged the incremental climbing task by more than 40 seconds, which cannot be attributed to better recovery or measurement error (Figure 3C). Therefore, some climbers seemed to be more affected by localized cooling than others. From this logic, it is not possible that a recovery strategy between trials (especially the short one such as 20 minutes) enhances subsequent performance but may only facilitate faster achievement of prefatigue level. Hence, it is questionable if the CWI during local isometric contractions facilitated recovery. It is possible that CWI may inhibit through thermoreceptors afferent feedback from the muscle (22) and therefore enhance subsequent muscle performance, but this would occur similarly in both precooling and mid-cooling conditions. From a physiological perspective, cooling muscle may hinder perfusion and O_2 delivery in the muscle (14,23) and actually slow down the recovery process. However, previous work has shown that faster pH normalization was observed up to 60 minutes after the exhaustive exercise (26). In that study, the pH increase was due probably to the inverse relationship between muscle temperature and pH proposed for humans to be -0.005 pH units/°C (27), but the exact mechanism was not described. We have not taken any physiological measures in the current study; however, our results do not show any additional benefits that could arise from a facilitated recovery, and the beneficial effects of cooling seem to be mainly explained by the “precooling effect.” If the mid-cooling should work as a recovery strategy, there should be

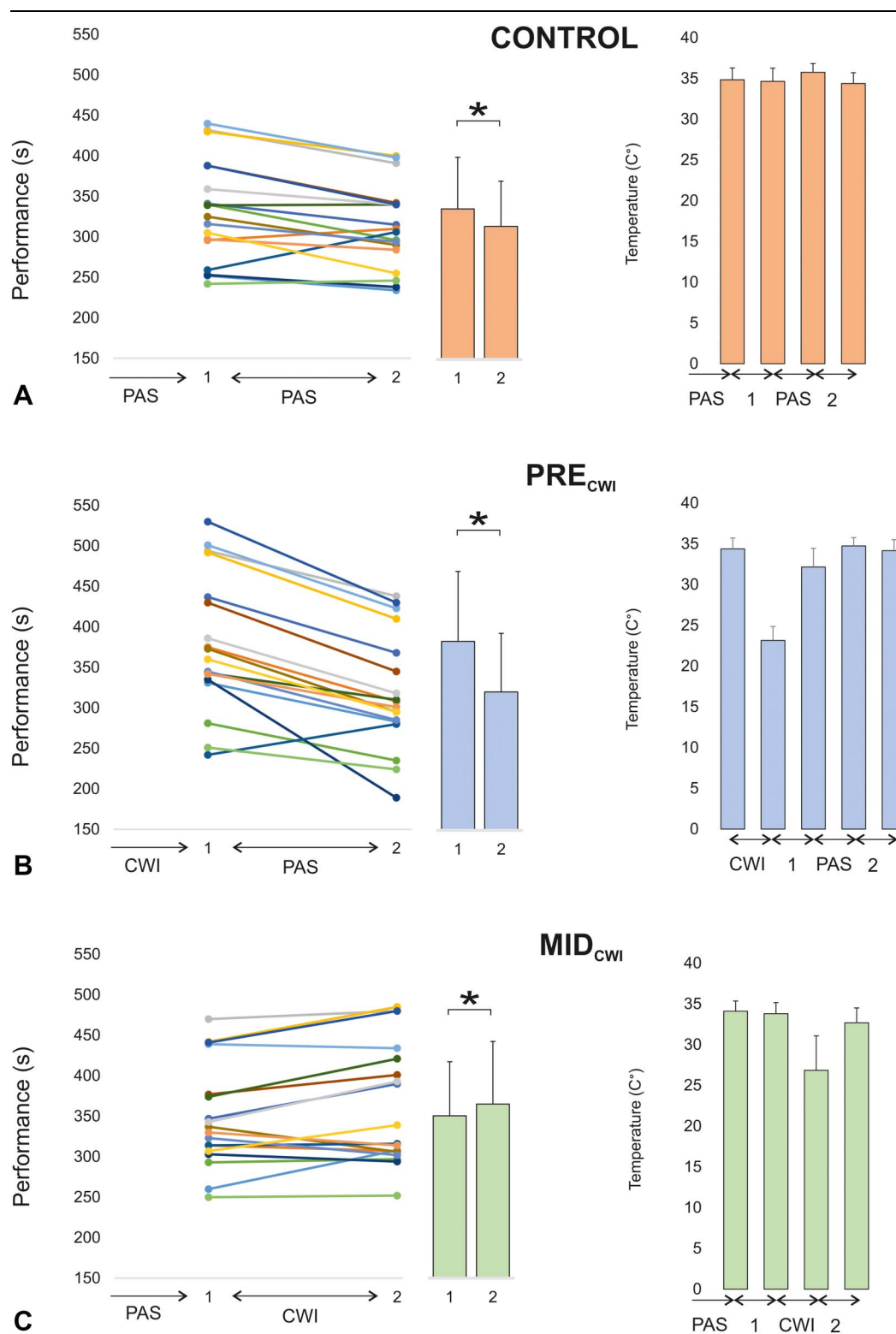


Figure 3. The performance (climbing time in seconds) of 2 repeated trials in 3 conditions: (A) CONTROL: 20 minutes of passive sitting (PAS) before the first trial and the second trial, respectively; (B) precooling (PRE_{CWI}): forearm cold-water immersion (CWI) before the first trial and PAS before the second trial, respectively; and (C) mid-cooling (MID_{CWI}): PAS before the first trial and 20 minutes of CWI before the second trial. Surface skin temperatures of the forearms immediately taken before and after climbing trials are depicted at the right part of the figure. *Significant time differences between first and second climbing trials.

“additional” benefits next to the precooling “stimulation” effect. Therefore, the time of performance enhancement with respect to PAS should be longer after mid-cooling than after precooling, which was not confirmed in the current study. Therefore, it is plausible that using CWI in a cooler environment whereby muscle temperatures are already lowered would not bring any additional benefits, as seen with higher muscle temperatures.

During whole-body exercise that heavily relies on local endurance such as sport climbing, the beneficial effect of muscle cooling seems to be mitigated when compared with isolated contractions (2,16). For instance, previous research showed that all male subjects improved handgrip performance by an average of 44% after forearm CWI (2). In the current study, we used the same CWI protocol and observed only a 4% (nonsignificant) climbing performance increase, and not all subjects improved (Figure 3A). Similarly, during exhaustive climbing at a self-selected speed, female climbers of another study completed an average of 3 hand moves (+2%) further after CWI with respect to pretest (150 vs. 147 moves, although not significant) (13). The mitigating effect of cooling during whole-body climbing (with respect to localized contractions) on muscle performance may be linked to lower dependence on glycolysis (total time of climbing was longer, likely meaning that the relative intensity was lower than during isolated handgrip contractions), interindividual impact of CWI (Figure 3A), or that whole-body task requires high level of coordination, which may be deteriorated with progressive fatigue.

Using CWI as a recovery strategy has been discussed in many narrative or meta-analytical reviews with controversial findings (15,19,21,24). However, existing review articles about the effects of CWI summarize data from very heterogeneous study designs, external conditions, and research samples (different CWI protocols—duration, temperature, whole-body or local CWI, different activities, types of contraction, intensity of exercise, various ambient temperature, different muscles, skinfold thickness, sex, etc). The current study applied a previously published CWI protocol that improved localized handgrip contractions to sport-specific setting, where localized contractions are the main fatigue factor. Although physiological measures such as heart rate, blood lactate, or local oxygen kinetics were not measured, the design of the study enabled to estimate the effect of performance recovery from the “precooling” effect. Apart from not having internal response from the recovery strategies, there are some more limitations to be stated. The study was completed only with male athletes. The response from female or nonathlete population is questionable and should be verified in future research. Moreover, to better understand the effect of repeated cooling, one condition cooling before and in the middle of a session would have been worth including in the study design; however, it was unfortunately not possible because of time restrictions.

To conclude, muscle precooling and mid-cooling prolong localized endurance performance during climbing in a warm environment. However, the effectiveness of mid-cooling as a recovery strategy may be attributed not necessarily of cooling an already overheated muscle but to a low muscle tissue temperature before subsequent performance (precooling, but in the middle of a session), which delays the onset of localized heat-induced neuromuscular fatigue. This study demonstrated that localized, and not systemic, factors during cooling have a positive effect on subsequent localized muscular endurance performance in a warm environment.

Practical Applications

In sports relying on forearm capacity to endure against fatigue, such as sport climbing, canoe-slalom, or combat sports, male athletes may benefit from localized CWI precooling and mid-cooling to enhance the sport performance in a warm environment. In practice, 3 cycles of 4 minutes of forearm immersion interspersed by 2 minutes of passive recovery at 15° C water prolong exhaustive sport climbing performance lasting from ~200 to 500 seconds by 9–14% on average. Shorter CWI protocols with cooler water (5) may provide similar benefits if they lead to optimal muscle temperature of ~27° C for localized muscle endurance performance (8). Precooling effect only lasts for the period of lowered muscle or skin temperature. With the skin temperature returned back to ~35° C, the beneficial effect of CWI is not to be expected. If the performance is undertaken in cool environmental conditions leading to lower muscle temperatures, the effect of localized CWI as a recovery strategy will probably be mitigated and other forms of recovery strategies (e.g., active recovery) from exhaustive contractions may be applied.

Acknowledgments

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